Compact and Bidirectional QKD BB84 Circuit on Photonic IC

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***Abstract*—A compact, bidirectional Quantum Key Distribution (QKD) BB84 circuit has been designed on a 220 nm Silicon-on-Insulator (SOI) photonic integrated circuit. The architecture, consisting of a single balanced 2x2 Mach-Zehnder Modulator (MZM) coupled to a 2D Polarization-Splitting Grating Coupler (2D PSGC), functions as both the transmitter (Alice) and the receiver (Bob) by leveraging the principle of optical reciprocity. The MZM phase shifter is segmented into length ratios of 1/3 and 2/3, which allows for separate bit and basis modulation using Non-Return-to-Zero (NRZ) signals. Validated via Lumerical INTERCONNECT, the Alice circuit achieves highly accurate polarization generation with a worst-case deviation of only 1.2 degrees. The transmission spectrum demonstrates an Extinction Ratio (ER) greater than 35 dB with an Insertion Loss (IL) of approximately 5.56 dB. On the receiver side, the system effectively distinguishes polarizations with an ER exceeding 45 dB and an IL of 4.48 dB. Under mismatched basis conditions, the identical signal waveforms yield an ER of less than 0.1 dB, rendering bits indistinguishable and confirming the expected key-shifting behavior. This MZM and 2D PSGC-based architecture offers a significantly compact footprint with precise polarization generation for reliable QKD applications.**

***Index Terms*—PIC, BB84, Segmented MZM, 2D PSGC**

I. Introduction

The rapid development of quantum computing to conventional cryptography necessitates Quantum Key Distribution (QKD), which ensures unconditional security via protocols like BB84. While silicon Photonic Integrated Circuits (PIC) offer a scalable platform for QKD, current implementations face architectural drawbacks: time-bin encoding requires bulky, thermally sensitive delay line [1], while on-chip dual-mode polarization encoding suffers from index differences and complex, lossy fiber coupling. To overcome these limitations, this paper proposes a compact, purely TE-based BB84 architecture utilizing a single Mach-Zehnder Modulator (MZM) and a two-dimensional Polarization Splitting Gratting Coupler (2D PSGC). By leveraging the MZM for high-speed, thermally stable modulation and the 2D PSGC to directly map TE phase states into fiber polarization states, this design avoids TE/TM dispersion and elongated couplers entirely, achieving a highly efficient, small size solution for secure QKD transmission.

II. Proposed Method

The proposed architecture uses a single 2x2 MZM as the primary modulator. While MZMs are typically employed as optical switches to toggle between all-through and all-drop output states [2], they can also be configured to produce a 50:50 ouput split, a technique previously demostrated for PAM4 modulation [3], [4]. This design further exploits the inherent phase differences of this 50:50 split across distinct voltage levels.

Consequently, the MZM will require four voltage levels to encode both the bits and bases into the four polarization states. However, because of the nonlinearity of the MZM, the four voltage levels are not uniformly spaced, which would make it not compatible with PAM4 signals. To mitigate this, our design uses segmented MZM which cuts the phase shifter into $2L/3$ and $L/3$ sections, with L being the total phase shifter length. Because phase accumulates linearly with distance, we can divide from the total $3\pi/2$ phase shift into π and $\pi/2$ phase shift. This way, each section can be driven with a simple NRZ signal, although will require higher modulation voltage than the usual V_π voltage.

The output of this MZM is coupled to a 2D PSGC to generate the four polarization states. This grating coupler works by superimposing two 1D grating coupler aligned orthogonally [5]. Because of this alignment, the final polarization on the fiber depends on the relative phase and power distribution of the input waveguides. For the rectilinear basis, horizontal and vertical polarizations, denoted as $|H\rangle$ and $|V\rangle$, are obtained by routing the light exclusively into one of the two waveguide arms. Conversely, the diagonal basis polarizations, denoted as diagonal $|D\rangle$ $(+45°)$ and anti-diagonal $|A\rangle$ $(-45°)$, are generated by routing the light into both arms simultaneously with equal power. The block diagram for this architecture can be seen in figure 1.

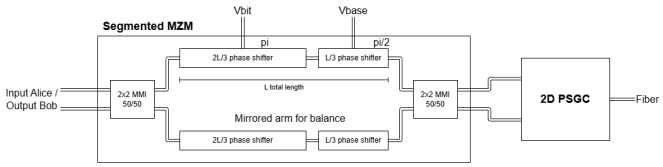


Fig. 1. Block diagram of the proposed BB84 circuit architecture.

III. Design and Simulation

The full circuit simulations were performed using commercial Lumerical INTERCONNECT software with 220 nm SiEPIC Ebeam PDK. As a proof of concept, the phase shifters in the MZM utilize the `wg_heater` component provided by the PDK. Based on simulation data, a $300\mu m$ shifter requires

a driving voltage of $1.2V$ to achieve a $3\pi/2$ phase shift, while maintaining a low 0.23 dB loss. The 2D PSGC component, however, had to be redesigned for a 220 nm SOI with 130 nm etch depth. This design uses corner mirrors to obtain a higher coupling efficiency [6]. After some optimization, an acceptable 38.3% coupling efficiency was achieved with a very low PDL of below 0.1 dB.

For Alice simulation, the circuit was characterized by varying the voltage levels for both the bit and base shifter, with $0V$ as logic level low and $1.2V$ as logic level high. The optical fiber output was then measured for C band which can be seen in figure 2. The transmission spectrum results indicate that the Alice circuit achieves an ER of greater than 35 dB (45 dB and 37 dB) for the rectilinear base and less than 0.4 dB (0.20 dB and 0.36 dB) for the diagonal base. Furthermore, the generated polarization angles are highly accurate, exhibiting a worst-case deviation of only 1.2 degrees ($-0.31°$, $89.2°$, $-45.6°$, and $43.8°$). However, the insertion loss remains relatively high at approximately 5.56 dB.

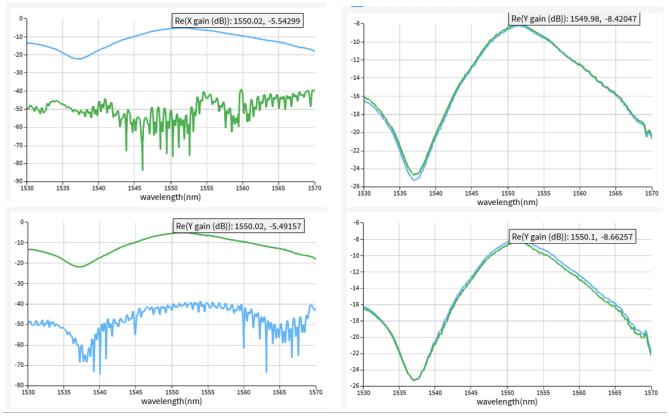


Fig. 2. Transmission plot for transmitter (Alice) simulation from Lumerical INTERCONNECT.

On the receiver side, the circuit was simulated by varying the input polarization using an ideal Polarization Rotator to the optical fiber port, while driving the correct voltage levels on the base shifter ($0V$ for rectilinear, $1.2V$ for diagonal). This simulation demonstrates that the circuit successfully distinguish the four polarization variations, achieving a very high ER exceeding 45 dB, albeit with a relatively high loss of 4.48 dB.

To validate the PIC architectures, we evaluate the system's performance under the asymptotic BB84 protocol using the standard GLLP security framework. The model incorporates our extracted worst-case hardware metrics: a total combined insertion loss (IL_{total}) of 10.04 dB, a transmitter extinction ratio (ER) of 37 dB, and a polarization rotation error (θ) of $1.2°$.

The simulated scaling of the Quantum Bit Error Rate (QBER) and Secret Key Rate (SKR) over a standard fiber channel ($\alpha = 0.2$ dB/km) using 20%-efficient InGaAs detectors is presented in figure 3. The system achieves an exceptionally low baseline QBER of 0.09% (158.56 kbps SKR) at 0 km, rising to 2.46% (1.15 kbps SKR) at 100 km, with a maximum secure range of 137.2 km under these device-limited conditions. Stress-testing against up to $10°$ of uncompensated fiber polarization drift narrows this only modestly to 130.1 km. Notably, this is achieved despite the PSGC's coupling efficiency of only 38%, well below the 70% reported for optimized design [6], underscoring that this architecture is an early-stage proposal with substantial headroom for further improvement.

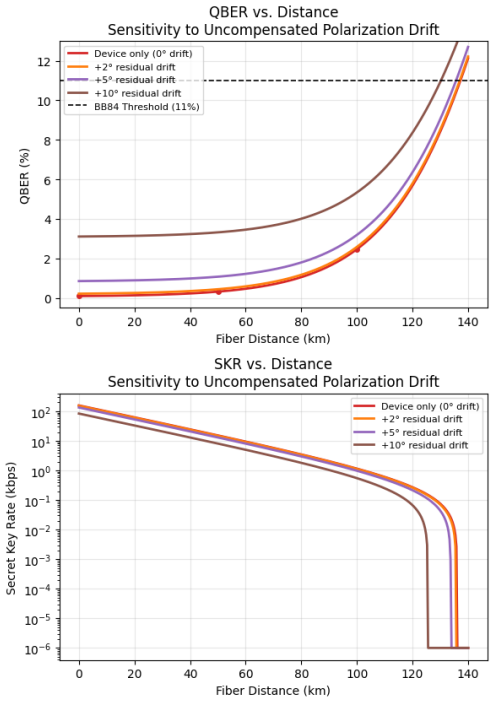


Fig. 3. Proposed design estimated metrics (QBER and SKER) over distance with polarization drift variations.

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